



Basic Details of the Team and Problem Statement



PS Code: **210**

- Problem Statement Title: **IOT based Water Quality Monitoring.**

Team Name: **JalNetra (जलनेत्र)**

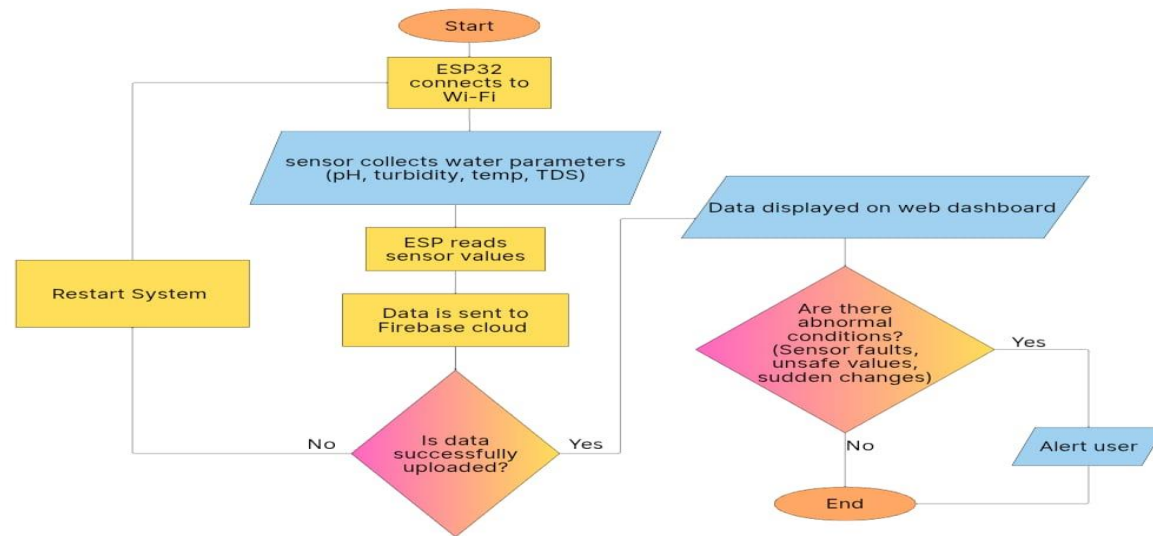
Team Leader Name: Sumit Singh

- Institute Name: **Lalbhai Dalpatbhai College of Engineering.**

Category Number and Name: **Category-2: Engineering and Technology.**



WORK FLOW



Idea/Approach Details

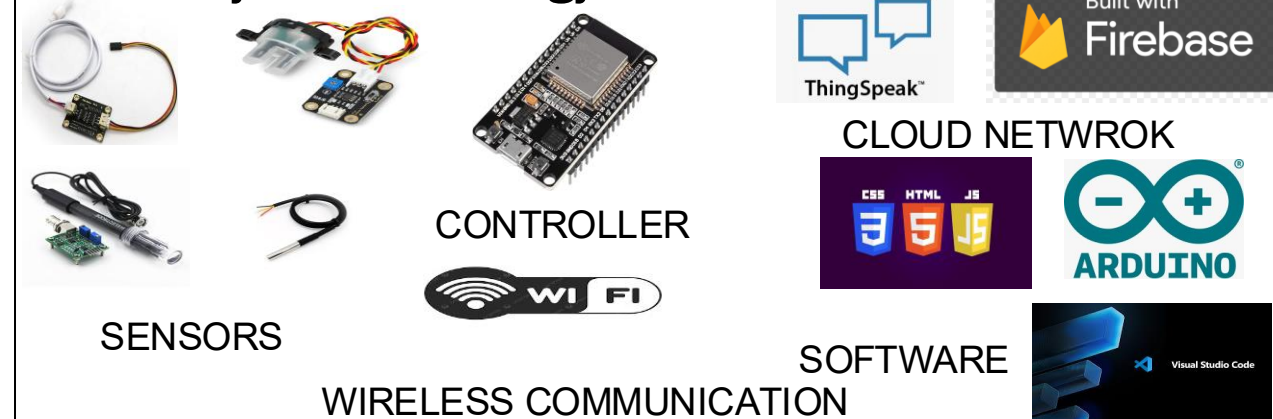
Describe your Idea/Solution/Prototype here:

- We propose a **Smart Water Monitoring System** using ESP32/Raspberry Pi and IoT sensors to collect real-time water parameters and store them on Firebase/ThingSpeak Cloud for remote monitoring and future analysis. The prototype focuses on data acquisition, cloud storage, and visualization, with **scope for predictive Machine Learning** as a future enhancement.

Prototype will function as follows:

- Water parameters such as pH, turbidity, temperature and status are collected using sensors connected to **ESP32/Raspberry Pi**.
- ESP32/ Raspberry Pi transmits **real-time sensor data** to **Firebase/ ThingSpeak Database** using Wi-Fi.
- Cloud data is displayed on a screen for real-time monitoring and also viewed with the past records.
- The system allows **remote access**, allowing users to **check water conditions** anytime, from anywhere.
- **Threshold-based alerts** can be added to detect abnormal conditions (sensor faults, unsafe water parameters, sudden value changes)
- Collected Firebase/ Thinkspeak data will later be used for predictive machine Learning to **forecast trends** and **improve decision making** (future scope)

Describe your Technology stack here:





Idea/Approach Details

Describe your Use Cases here

1. Environmental & Ecosystem Monitoring

- Regulatory agencies can deploy the system along rivers and lakes to detect **pollution spikes** caused by industrial discharge or sewage overflow, helping track overall **water health**.

2. Early Warning & Public Health Safeguards

- **Real-time alerts triggered** by sudden parameter spikes indicating possible contamination.
- Example: A lake system sends alerts when turbidity rises after heavy rainfall, indicating stormwater inflow carrying sediments and contaminants.

3. Urban Water Infrastructure & Smart Cities

- **Monitoring** stormwater drains, reservoirs, and canals.
- Integration with **city IoT platforms** allows municipalities to respond quickly to water quality threats.

4. Industrial Compliance & Effluent Monitoring

Industries can monitor wastewater discharge to ensure:

- pH stays within **legal limits**
- Temperature remains in **allowed range**
- Measures can be taken to Control Turbidity/Solids

#Future Trends:-

- **AI & Predictive Modeling** – Predict pollution trends before critical levels.
- **Low-Power IoT & Edge Computing** – Reduce cloud dependency and improve battery life.

Describe your Dependencies / Show stopper here

1. Dependencies:

- Stable Wi-Fi for real-time cloud updates.
- Periodic sensor calibration for accuracy.
- Reliable power source (battery/charging).
- Proper waterproof enclosure for field deployment.
- Standard reference values for safe/unsafe classification.

2. Stopper:

- Sensor fouling due to continuous water exposure.
- Drift in pH and TDS sensors over time.
- Limited detection of heavy metals and toxic compounds.
- Internet unavailability in remote areas.
- Extreme environmental conditions affecting electronics.



Team Member Details

Team Leader Name: Sumit Singh		
Branch: B.Tech	Stream: Instrumentation & Control Engineering	Year: II
Team Member 1 Name: Khushi Mehta		
Branch: B.Tech	Stream: Instrumentation & Control Engineering	Year: II
Team Member 2 Name: Brihati Vyas		
Branch: B.Tech	Stream: Instrumentation & Control Engineering	Year: II
Team Member 3 Name: Mannan Tank		
Branch: B.Tech	Stream: Instrumentation & Control Engineering	Year: II
Team Mentor Name: Dr. Kunal Bhoyania		
Category: Academics	Expertise: Instrumentation Systems	Domain Experience: 25+ Years